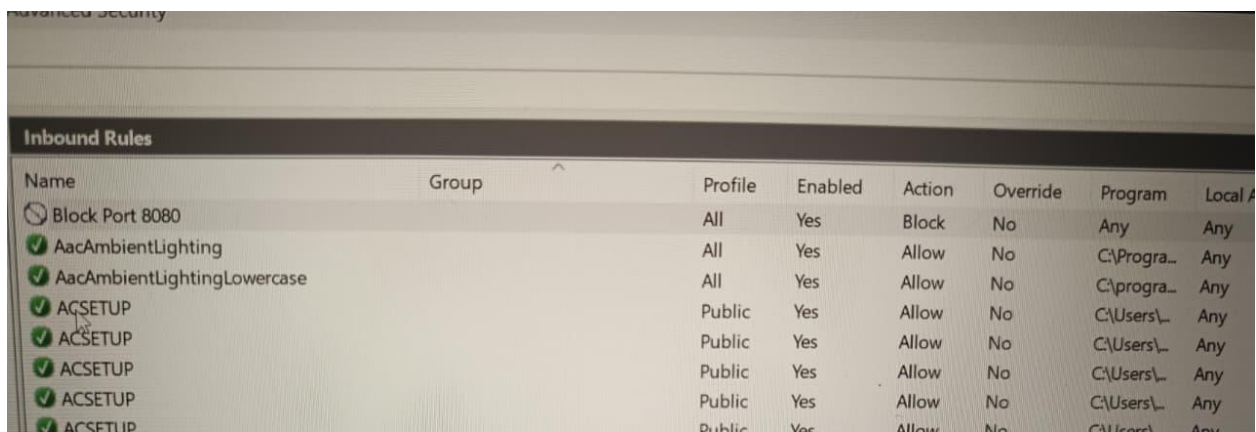


SECURE NETWORKS WITH FIREWALLS NAT AND OTHER METHODS

1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

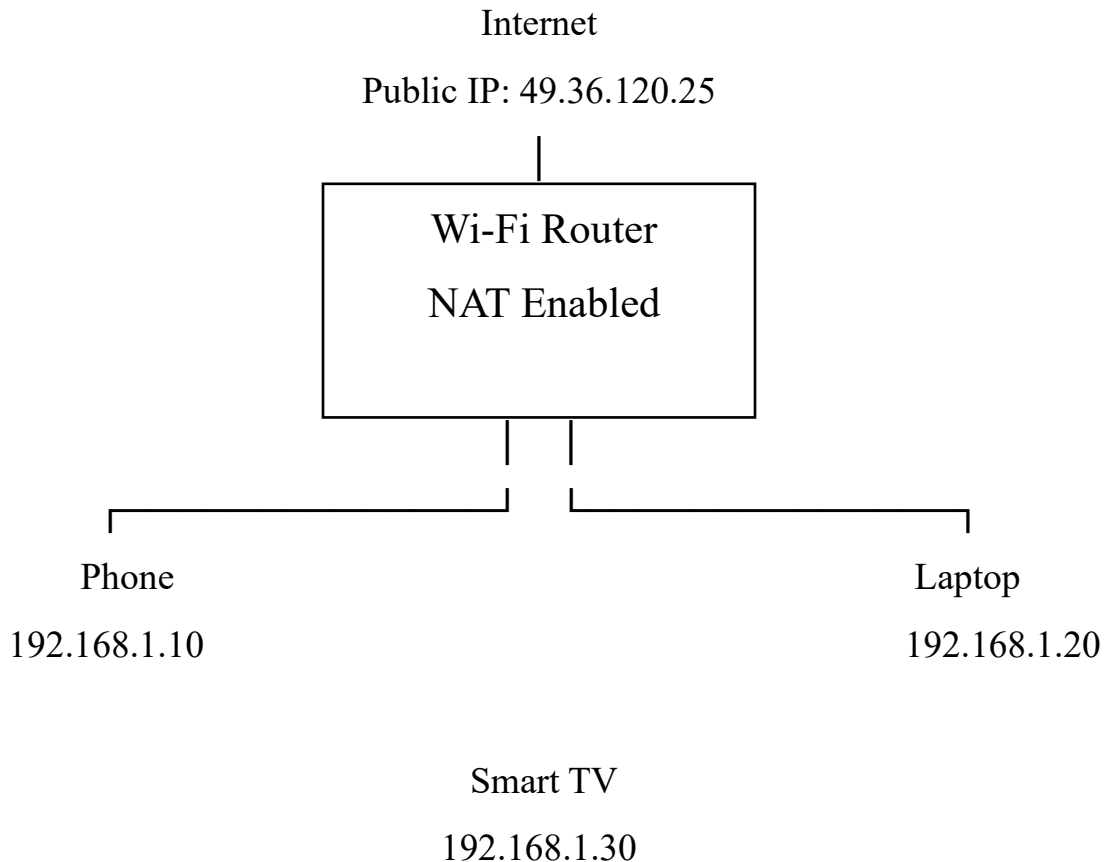
Steps

1. Open Windows Defender Firewall with Advanced Security.
2. Click Inbound Rules → New Rule.
3. Select Port → Next.
4. Select TCP and enter 8080.
5. Select Block the connection.
6. Apply the rule to Domain, Private, and Public profiles.
7. Name the rule Block Port 8080 and click Finish.



**2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

Hint: Use any drawing tool or even hand-draw and click a photo.**



3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Steps

1. Log in to your router's admin panel.
2. Open Firewall or Access Control settings.

3. Create a rule to Block Outgoing TCP Port 25.
4. Save and apply the settings.

Explanation

- Blocking port 25 (SMTP) prevents devices from sending unauthorized email directly.
- This helps reduce spam and prevents malware from using your network to send spam emails

**4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Hint: Focus on source/destination IPs and ports in the rule set.**

Note: The answer depends on the rule set provided by your trainer. Without the rule set, it is not possible to determine whether these packets are allowed or blocked.

Sample packets:

1. HTTP: 192.168.1.5 → 172.217.166.78:80
2. SSH: 10.0.0.10 → 34.201.23.1:22
3. DNS: 192.168.1.7 → 8.8.8.8:53

Upload or share the packet filtering rule set, and I can tell you which packets are Allowed or Blocked

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

- Block Unauthorized Access: Firewalls block hackers from accessing servers through unused or dangerous ports.

- Allow Only Required Traffic: Packet filtering allows only trusted traffic (such as HTTPS on port 443) and blocks unwanted connections.
- Protect Against Attacks: Firewalls detect and block suspicious traffic such as DDoS attacks or repeated malicious login attempts, helping keep the service available for users